1) _ _ _ _ _

{0, 1, 2, 3, 4, 6, 8}

Case 1: $\overset{\neq 0}{\underset{\downarrow}{_}}\ \underset{\downarrow}{_}\ \underset{\downarrow}{_}\ \underset{\downarrow}{_}\ \underset{\downarrow}{\overset{0}{_}}$ $\Rightarrow 30 \cdot 12 = 360$

$\Rightarrow 1960$ nr

$\underset{\downarrow}{_}\ \underset{\downarrow}{_}\ \underset{\downarrow}{_}\ \underset{\downarrow}{_}\ \underset{\downarrow}{\overset{\neq 0}{_}}$ $= 64 \cdot 25 = 1600$

2) _ _ _ _ {1, 2, 3, 4, 5}

a) $\underset{4}{_}\ \underset{4}{_}\ \underset{4}{_}\ \underset{4}{_}$

$\Rightarrow 256$

$\underset{3}{_}\ \underset{3}{_}\ \underset{3}{_}\ \underset{3}{_} = 81$ $\Rightarrow 256 - 81 = 175$

b) {1, 2, 3, 4, 5, 6} $\underset{5}{_}\ \underset{5}{_}\ \underset{5}{_}\ \underset{5}{_} = 625$

$\underset{6}{_}\ \underset{6}{_}\ \underset{6}{_}\ \underset{6}{_} = 1296$

$\Rightarrow 671$ nr

c) $\underset{6}{_}\ \underset{7}{_}\ \underset{7}{_}\ \underset{7}{_} = 2058$

$\underset{5}{_}\ \underset{6}{_}\ \underset{6}{_}\ \underset{6}{_} = 1080$

$\Rightarrow 978$ nr

{0, 1, 2, 3, 4, 5, 6}

3) $A=\{1,2,\dots,10\}$ $\{1\}\to\{2,\dots,10\}$

a) $2^{10}\cdot 2 = \frac{1022}{2} = 511$

$3,\{4\},\dots,10 = 2^8 = 256\cdot 2 = 512$

$\{1,2,3,4\}$ $3,4,\{3,\emptyset\}$

$3\to 4\to 2$ $3\to 4$

$\emptyset\to 3,4\to 2$ $4\to 3$

$3,4\to\emptyset$

$3\to$ $\emptyset\to 3,4$

$\frac{2^8}{2}\cdot 2 = 2^8$

b) $(1,2)\to\{3,4,5,\dots,10$ 2^8

$511-256=255$

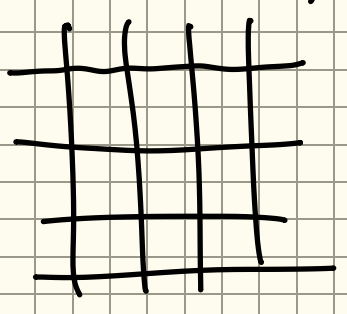
5) b) $8^2 = \frac{8\cdot 7\cdot 6}{6} = 8\cdot 7 = 56$

$56\cdot 170 = 6170$

$5! = 120$

a) $56\cdot 2\cdot 120 = 13440$

6) a) $8!$



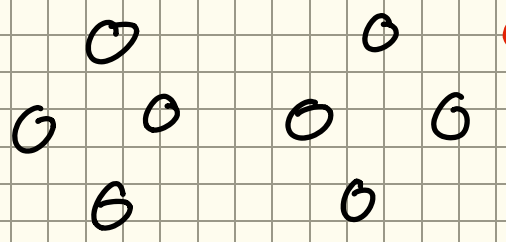
?

b) $8^{\underline{2}} = 8 \cdot 7 \cdot 6 = 56 \cdot 6 = 336$

$\frac{336}{6} = 56$

c)

7)



$8^{\underline{4}} = 8 \cdot 7 \cdot 6 \cdot 5 =$

8)



10) $1 \to 2$ $\quad 1 \to 5$ $\quad 15$

$2 \to 1$

$1 \to 2, 3, 4, 5, 6$ $\quad 15$

$1 \to 5$

$2 \to$

$6^{\underline{6}} = 6 \cdot 15$

$f(1) = 2$

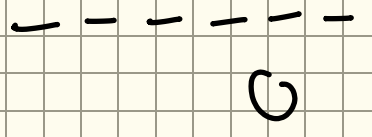
$f(2) = 3$

$f(3) = 4$

$f(5) = 6$

$f(4) = 5$

$f(6) = 1$



5) $1, 2, \ldots, 8$

312
213

$1, 2, 3$

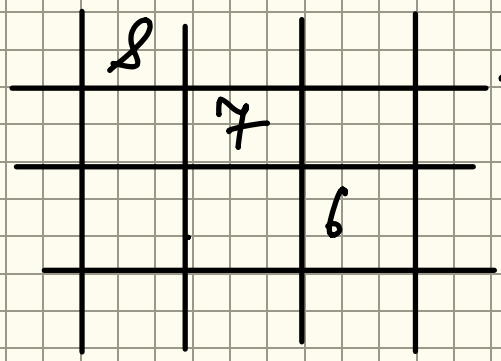


$$C_8^3 = \frac{8!}{5!\,3!} = \frac{8 \cdot 7 \cdot 6}{6} = 56$$

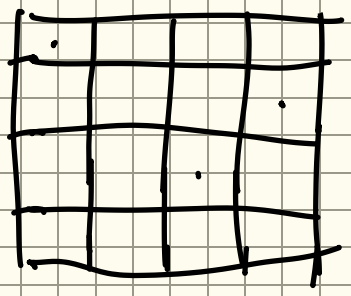
$$56 \cdot 2 \cdot 120 = 13440$$

$$5! = 120$$

6)

a) 

$= 8!$

b) 

$1 \quad 2 \quad 3$

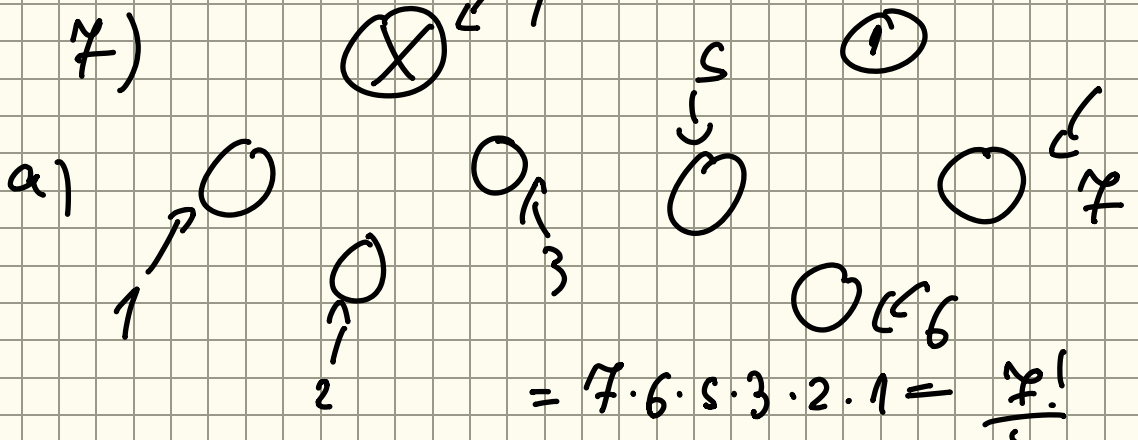
$C_8^3 = 56$

$56 \cdot 56 \cdot 3 = ?$

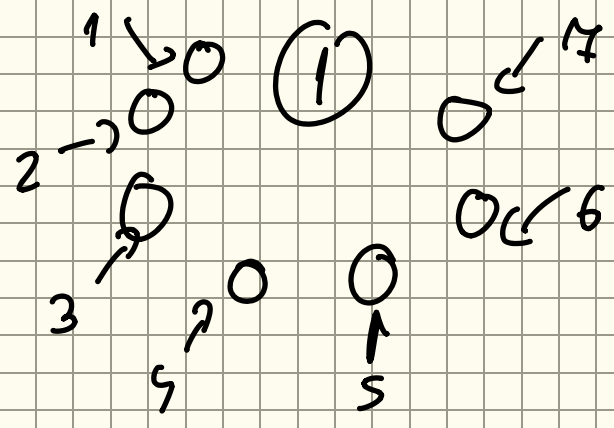
c) 56^2

d) $\left(\frac{n!}{k!(n-k)!}\right)^2$

7)

a) 

$= 7 \cdot 6 \cdot 5 \cdot 3 \cdot 2 \cdot 1 = \frac{7!}{4}$

b) 

7! =

8) a) 3! -> perm A, B, C

A ⊗ B ⊗ C

220 · 3!

14 − 5 = 9

12 $C_{12}^3 = \frac{12!}{9!3!} = \frac{10 \cdot 11 \cdot 12}{6} = 220$

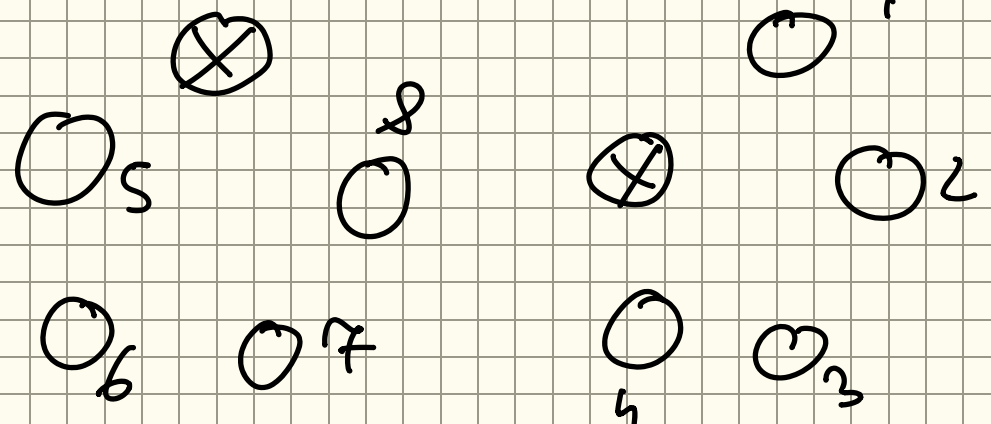
b) 3! - A, B, C

A x x B x x C

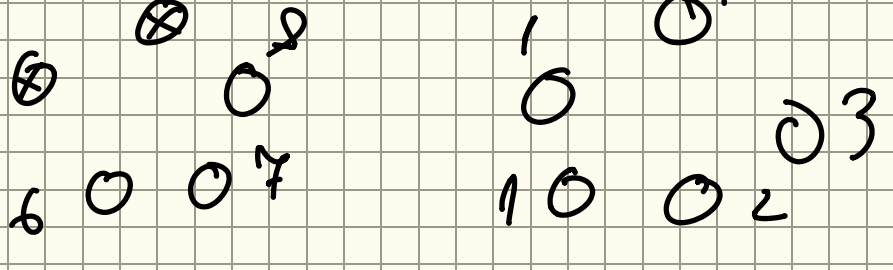
C

14 − 4 = 10

?? — — — — — — — — — — — — — —

9) 

8! +



$\frac{8!}{5} + 8! =$

